HTE operator handbook

Local OT-2 dual-ligand fluorination screen

Edition: 16 October 2026 (Europe/Zurich). Generated from editable repository sources.

HTE configuration review - bench checks remain open

Campaign	48 first-round wells; 23 second-round wells
Reaction	25 µmol substrate; 250 µL toluene; 95 °C; 240 min per round
Catalysts	6 mol% each ligand; 12 mol% Pd(COD)(DQ)
Sample dilution	20 µL extract + 980 µL IS-free LC diluent
Software evidence	145 local tests; 0 failures. Initial Claude scores are real; practice chemistry and loop responses are synthetic.
Bench readiness	12 applicable validation fields still unconfirmed; exact hardware and analytical settings must be supplied by HTE.
Provisional timing	717.7 min target with staged LC; 752.7 min serial / 720 min limit. Handling and full LC cycle are unmeasured.

How to keep this handbook current

Edit the Markdown sources in **docs/** and the relevant campaign configuration, then regenerate locally or download the fresh GitHub workflow edition. The PDF is a dated export; the editable sources remain the master. Keep the PDF with its build manifest to check for later source changes.

```
python scripts/build-hte-pdf.py
```

Open the shared repository - see PDF update instructions.

Generated quantities from the real selected campaign take precedence over the synthetic demo. This handbook includes the procedures and outstanding decisions; it does not supply unmeasured geometry or analytical settings.

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1. Quickstart and responsibilities

Editable source: docs/start-here.md

Use the two-page staff checklist at the robot. This page is for the colleague running the local optimizer. The initial **48-pair design is already prepared** in `campaigns/hackathon2026/round1_design.csv`. Do not rerun paid initial scoring unless deliberately making a new campaign.

Local setup before the timed run

Run commands from the repository root. Use Python 3.12; HTEL's normal `uv sync --python 3.12 --extra sources` and `uv run` are supported. Alternatively install the project with `python -m pip install -e '.[sources]'` inside an activated virtual environment.

```
mkdir -p local_data
cp hte_inputs/campaign.json local_data/campaign.json
cp hte_inputs/calibration_template.csv local_data/calibration-plan.csv
cp anthropic.key.example anthropic.key
```

Do not overwrite an existing key. Open `anthropic.key`, replace its single placeholder line with the raw Claude key, save, and restrict access with `chmod 600 anthropic.key`. Never paste it into a protocol, CSV, GitHub or an email. No robot credentials are required.

In `local_data/campaign.json`, record the actual 1.7-minute **method ID**, UV channel, authentic-standard retention times/windows and experimentally supported SM/product LOQs. Keep the requested 210 nm wavelength distinct from the channel name. Complete applicable `validation` fields only from local setup evidence. A separate extraction-homogeneity study is not required by the team; visible layers/solids still need review. Remaining facts.

```
python -m hte.cli --config local_data/campaign.json check
python -m hte.cli --config local_data/campaign.json export \
  --design campaigns/hackathon2026/round1_design.csv \
  --output output/r1-dose --live --stage dose_and_react
```

Import `output/r1-dose/dose_and_react.py` in the local App, simulate and start locally. Use its matching CSVs. If setup is incomplete, export without `--live` to inspect/simulate; never remove the `DEMO` guard manually. First dosing can be released before calibration finishes, but all other applicable checks and a known LC method are required.

During the first four-hour hold

Prepare/run the 14 authentic-standard injections and complete calibration import. Staff also predose/dry **23 SM-only vials** in an ordinary off-deck rack; no second reaction block is needed. After calibration passes, record `lc_calibration_validated=true` and export workup separately:

```
python -m hte.cli --config local_data/campaign.json export \
  --design campaigns/hackathon2026/round1_design.csv \
  --output output/r1-workup --live --stage workup
```

After all 48 first-round injections

Fill each generated `lc_batchNN_injections.csv` with the actual method/sequence identities. Keep one accepted measurement per sample. Six batches may reset injection numbering; each export needs its own map.

```
python scripts/hte-lc-results.py next \
  --dosing output/r1-dose/dosing.csv \
  --exports local_data/r1-batch*.csv \
  --injections local_data/r1-map*.csv \
  --output output/r1-decision
```

Both globs must expand to matching batch order and must not include unrelated files. Calibration defaults to `local_data/calibration.csv`; use `--calibration` if elsewhere. This command checks all 48 samples, calibration and QC, then calls Claude on the **387 untested pairs** in small requests. Both stages include the same single-ligand SI yields. It outputs **22 new pairs + one L17 single reference**. No embedding/GoLLuM selection runs here.

The printed decision directory contains `next_design.csv`, `next_dosing.csv`, analysis and score records. If all setup checks pass, its `protocols/` contains released local Python. Otherwise, complete the setup evidence and export explicitly using that `next_design.csv`, `--history output/r1-dose/dosing.csv`, a fresh output directory and `--live`. Use matching second-round deck loads; replace the first 48 vials with the 23 dry predosed vials. Do not redose SM.

After the 23 second-round injections, use the same helper with `final`, second-round `dosing.csv`, three matching LC exports/maps and a fresh output directory. It calculates results without calling Claude. Compare both rounds' calibrated pair yields with the measured L17 reference, preserving the different run times. One reference is a resource-efficient comparison, not proof of pair synergy or reproducibility.

Practice / recovery

`python -m hte.cli rehearse-loop --output output/practice` makes no paid calls and changes no robot. All practice yields/next pairs are synthetic. Use new output directories for corrected LC imports. Scoring can reuse validated responses from its exact original scoring directory, but an interrupted/uncertain request requires inspecting its attempt/raw files before any deliberate repayment. Never repeatedly relaunch a failed watcher event blindly.

2. Complete operator runbook

Editable source: docs/hte-runbook.md

The staff sheet is the action sequence. This reference explains the quantities, release steps and analytical handling. Use generated CSVs from the same export as the Python protocol; they are the authority for well identities and source loads.

Fixed experiment and comparison

Substrate: **pyridine-2-sulfonyl fluoride / PyFluor**, `FS(C1=NC=CC=C1)(=O)=O`. Product: **2-fluoropyridine**, `FC1=NC=CC=C1`. Each 250 µL reaction contains 25 µmol substrate in toluene, each ligand 1.5 µmol, Pd(COD)(DQ) 3 µmol and mesitylene 2.5 µmol. Hold at 95 °C for four hours under air, 500 rpm. No added base or external fluoride. Prepare ligand stocks in the glovebox and preserve capped reserves.

Round one is 48 distinct pairs covering all 30 available ligands. Round two is 22 untested pairs plus L17 tBuBrettPhos at **12 mol% total ligand**, otherwise identical. This gives a minimal measured single-ligand reference; without repeats or both corresponding single ligands, a higher pair yield supports improved performance relative to that reference, not a statistical or mechanistic proof of synergy.

Labware, pipettes and manual caps

Use the supplied 48 × 2 mL JSON and derived heater-stack definition in `hte_inputs/labware/`. Wells have **13 mm pitch**, so reaction transfers use p300 single GEN2 on the left. The p300 multi GEN2 on the right doses the standard 96-well LC preparation plate, not the reaction block. Standard footprint does not establish multichannel compatibility.

All vial pipetting happens in slot 3, away from the heater-shaker. Slot 10 has the universal flat adapter; slots 7 and 11 remain empty. The derived JSON preserves all supplied well coordinates and adds a zero-overlap flat stacking declaration and **45 mm capped travel envelope**. Local setup must establish secure mounting/latch closure and cap height within this envelope. The JSON/simulator cannot establish physical fit or temperature transfer. Opentrons adapters.

The OT-2 has no gripper. Every block move, cap opening/closing and tip-rack replacement is a local pause. Caps must be removed before aspiration and hand-tightened before reaction or high-speed workup shaking. During sampling, other vials stay capped; only the indicated batch is opened. No stir bars are used. Inspect actual source tubes and vial bottoms when setting offsets; defaults are **1.5 mm above bottom**, not bottom contact or assumed liquid-level tracking.

Dosing and overlapped preparation

Dose 50 µL of 500 mM SM in DCM. The protocol moves the block OFF_DECK at a pause for the facility's approved ventilated evaporation. Confirm dry SM before adding mesitylene or catalysts; residual DCM would alter the reaction solvent/volume. Return the cooled block to slot 3. Add 25 µL 100 mM mesitylene, 37.5 µL each 40 mM ligand, 75 µL 40 mM Pd, and 75 µL toluene: **250 µL total**. For L17 single reference, replace both ligand portions with 75 µL L17.

During the first four-hour hold, staff prepare/inject the 14 calibration samples and manually predose the 23 second-round vials with **SM only** in an ordinary off-deck rack. Evaporate, label and cap dry SM. Keep these separate from the active block. After first-round sampling, archive its 48 vials and reuse the one block with the 23 fresh predosed vials. Mesitylene is dosed once after dry-down in each round. Neither workup nor LC diluent contains additional IS.

Common stocks use the 12-channel Axygen reservoir; ligand stocks use ordinary plastic source racks in slots 1 and 4. Prepare bulk glass stocks separately and load only `deck_loads.csv` quantities. The stock table includes 50% working margin, dead volumes and conservative reserve for any second-round choice. Refill fresh stage sources; never pour recovered working liquid into bulk reserves.

Quench, mixing and representative sampling

Cool the sealed reaction block off the heater before opening. Add **500 µL 9:1 ACN/water +0.15% v/v formic acid**, giving nominal **750 µL extract, 0.1% acid** and 1.25 mL nominal headspace. Hand-recap every vial, then shake **1500 rpm for two minutes**, heater off, and return to slot 3. Ensure the module/block has cooled sufficiently for the intended workup temperature. High rpm mixes the liquid but does not establish chemical quench completeness or single-phase extraction.

The ACN-rich mixture is chosen to reduce the phase-separation concern from adding equal water/ACN to toluene. Separate formal recovery/homogeneity studies are waived by the user. Inspect visible appearance; persistent separate layers, solids at the aspiration path or bubbles require a pause, not selecting whichever layer is convenient. Do not filter/centrifuge/decant without documenting and assessing analyte/IS recovery. Toluene/ACN/water phase-equilibrium evidence.

Sample in batches of eight to overlap LC with preparation. Before each batch, gently invert/remix the indicated **capped** vials, let obvious bubbles settle and remove only those caps. Fresh robot sample tips perform two slow same-well prewet cycles before the measured aspiration, with no air blowout into the sample. Then aspirate 21 µL at 1.5 mm, retain 1 µL reverse-pipetting surplus, add a 2 µL air gap, dispense the nominal **20 µL sample** noncontact into **980 µL diluent**, and discard the tip. The source is never revisited after LC contact.

At the robot pause, recap the sampled reaction vials. Keep the preparation plate seated in slot 6. With **one fresh manual tip per completed LC sample**, mix 500 µL five times without drawing air, transfer 900 µL into its labelled HPLC screw-cap vial, cap and invert briefly. Keep maps explicit: reaction A1-F8 is different from LC A1-H6. Submit each completed batch before resuming the robot. Round two has 8/8/7 samples; H3 gets only diluent and is excluded from LC injections.

Solvent delivery and failed transfers

Condition fresh dosing tips twice with their own source; work from capped reserves and uncover active sources immediately before use. Slow aspiration, short settling delays, a small air gap and reverse surplus are starting settings for volatile organic solutions. Avoid immersing tips too deeply or touching walls/bottoms. A source must stay above the aspiration point after dead-volume allowance. Do not aspirate a suspension as though it were a known stock concentration; repeated aspiration can add bubbles without dissolving solids.

Tip reuse is restricted to consecutive deliveries of the **same source stock**, with destination dispensing above the top and no contact, mixing or destination aspiration. Different ligands and every analytical sample require fresh tips. If source/destination contact, dripping or unintended air occurs, pause, change the tip and record the affected sample. Refill according to source maps rather than guessing a top-up. For an uncertain analytical dilution, prepare a new dilution from freshly remixed original extract and record its mapping; do not rescue it by estimated liquid addition. Keep one accepted measurement per sample, with rejected measurements retained and a reason. Do not choose the larger yield from repeats.

Local processing and records

The start page gives initial release, LC helper and second-round export commands; LC import covers detector/method and calibration configuration. Robots remain islands: no API key or optimizer installation goes on the OT-2, and no script remotely starts it.

Archive source exports/maps, stock preparation/assay records, actual method and wavelength, calibration/checks, ready markers, all QC results, paid scoring responses/cache ledger, final CSVs, matching generated protocols and robot JSON logs. Keep experimental data private until the team authorizes publication. The handoff does not contain a Claude key.

3. Stocks and analytical calibration

Editable source: docs/stocks-and-calibration.md

Current master: `hte_inputs/campaign.json`, schema 2. Use the actual selected campaign's `stock_preparation.csv` for each ligand mass, bulk preparation volume and source address; `deck_loads.csv` supplies the different dosing/workup loads. **Preparation volume is not on-deck fill volume.** Masses below assume 100% assay; divide by the COA assay fraction and use appropriate solvent-compatible closures.

Reaction stocks

Reagent	Bulk preparation	Per reaction
SM, MW 161.16	**10.00 mL, 500 mM in DCM**: 805.8 mg, make to final volume; capped 20 mL glass vial	50 µL =25 µmol; evaporate ALL DCM
Each ligand	**40 mM in toluene**, glovebox; selected-campaign CSV contains individual quantities and reserves	37.5 µL =1.5 µmol each
Pd(COD)(DQ), MW 378.8	**40 mM in toluene**; R1 campaign stock plan prepares 9.53 mL, 144.399 mg	75 µL =3 µmol
Mesitylene, MW 120.19	**5.00 mL, 100 mM in toluene**: 60.095 mg	25 µL =2.5 µmol, AFTER dry-down
Neat toluene	At least generated reserve; current plan 15.57 mL	75 µL

Use ordinary source racks; fancy metal hardware is only the reaction block. Keep bulk ligand reserves in capped glass vials; load the configured working tubes only to their CSV quantities (maximum 1.4 mL). Each stock includes conservative second-round reserve and 50% margin. L17 reserve includes its single reference. All 30 active sources, including small-use ligands, appear on the deck-load sheet. No separate catalyst-solubility study is requested; an observed unsolved suspension still cannot represent a known dissolved concentration.

Quench and dilution solvent

- **Quench, 100 mL:** premix 9:1 ACN/water. Add **150 µL formic acid**, make to 100 mL with that mixture (0.15% v/v). No IS. Dose 500 µL into 250 µL reaction; recap and shake 1500 rpm for two minutes.
- **LC diluent, 125 mL:** premix 1:1 water/ACN. Add **125 µL formic acid**, make to 125 mL with that mixture (0.1% v/v). No IS. Dose 980 µL and add 20 µL extract.

Nominal final sample: **1.00 mL**, analyte maximum **666.667 µM**, mesitylene **66.667 µM**, formic acid **0.1%**, toluene **0.667%**. This is 64.7 µg/mL product or 107.4 µg/mL SM at 100% amount. The concentration is a reasonable starting load for LC-UV/MS, not an established detector range or an approved injection volume. For context, a Thermo LC-MS technical note uses 100 µg/mL preparations; that does not validate this method/analytes. Set the actual injection volume with the facility's existing 1.7-minute method and calibration.

Matched calibration stocks

Analyte concentration varies; **IS is constant**, including blanks/checks. Prepare separate SM and product standards, no reaction-catalyst mixture. A convenient nominal final matrix is **0.667% toluene, about 50.15% ACN, 49.08% water, 0.1% formic acid**. For 100 mL: 0.6667 mL toluene, 50.1492 mL ACN, 0.1000 mL formic acid, water to volume. This approximates the planned final diluted extract and ignores mixing-volume contraction. Prepare all analytical stocks in this same matrix, so adding them preserves solvent/acid proportions.

For each authentic analyte, prepare **10 mM, 10.00 mL**: SM **16.116 mg** or product **9.709 mg**, make to volume with matched matrix. Prepare **1 mM working** by diluting 100 µL of 10 mM to 1.00 mL. These analyte masters contain **NO mesitylene**. Prepare separate independently weighed analyte masters for the check standards; a second dilution of the same master does not check that master's preparation.

Prepare analytical mesitylene **5 mM, 20.00 mL**: **12.019 mg**, make to volume with matched matrix. This is separate from the 100 mM reaction mesitylene stock; using that toluene stock directly would change matrix proportions.

Fourteen calibration injections during round one

Prepare the following **1.00 mL** samples for each analyte. Add **13.333 µL of 5 mM analytical mesitylene to every sample**, then the analyte aliquot and matrix to final volume. Use suitable calibrated manual pipettes. Cap, mix and label with `calibration_template.csv` IDs.

Target analyte, µM	Role	Analyte stock, mM	Analyte aliquot, µL	Matrix, µL
0	Fit blank	None	0	986.667
10	Fit	1	10	976.667
50	Fit	1	50	936.667
150	Fit	10	15	971.667
350	Fit	10	35	951.667
800	Fit	10	80	906.667
666.667	Independent check	10, independent master	66.667	920.000

This brackets the nominal 0-100% SM/product range. Fit the blank plus five nonzero levels; keep the independent check outside the fit. Check linear response, separation from mesitylene/other compounds, and practical sensitivity on the actual detector. The lowest standard is not automatically a measured LOQ. Current software requires R^2 >=0.995 and independent check error <=15%; retain failed measurements rather than changing results to pass.

SM/product quantities and fixed IS are entered in the calibration plan; the importer extracts peak areas. If Chromeleon omits a verified zero peak, supply an explicitly reviewed integrated zero measurement; the software does not silently invent one. A positive omitted peak is not zero. No actual calibration or retention times are prefilled in the public campaign.

4. Facts to confirm with HTE

Editable source: docs/hte-questions.md

The campaign now uses the supplied **48 × 2 mL Para-Dox**, OT-2 heater-shaker, no stir bars, manual screw caps, 250 µL reactions and 48 +23 vials. These decisions supersede the earlier 24 ×4 mL / external-stirrer discussion.

1. Confirm the supplied block fits securely on the universal flat adapter/latch with all caps, the capped assembly stays within the **45 mm travel envelope**, and 95 °C/500 rpm reaction plus 1500 rpm workup are workable. Record any thermal equilibration time and actual cooling time; revise the timing budget accordingly. Pipetting is off-module in slot 3.
2. Supply the **existing 1.7-minute method ID**, full injection-to-injection time including washes/re-equilibration, injection volume and actual UV channel/wavelength. We request 210 nm initially. Record SM, 2-fluoropyridine and mesitylene RT windows/LOQs from authentic standards. The old example's 253/270/300/370 nm settings are not our method settings.
3. Explicitly schedule the **14 calibration/check injections during the first hold**. Confirm that batches of eight HPLC vials can begin injection while the robot prepares subsequent batches. The 12-hour plan needs 35 minutes of this overlap and has only about two minutes spare under its current unmeasured handling assumptions. Confirm a realistic timed setup/handling plan before release.
4. Mount p300 single GEN2 left and p300 multi GEN2 right; check custom labware offsets and source/sample immersion above the bottoms. Confirm the usual solvent/evaporation handling, capped dry-SM storage during the first hold and short-term source-container compatibility. Stock solutions use ordinary racks; bulk reserves stay capped.

Method/deck facts go in `local_data/campaign.json`; record local evidence before changing release flags. No separate solubility or extraction-homogeneity study is requested. Visible transfer faults/layers/solids are addressed at the protocol pauses. The pipeline, key file and LC processing stay on the local workstation; only released Python/CSV setup files go to the OT-2.

5. Claude prompt caching and costs

Editable source: docs/prompt-caching.md

The live initial campaign used **claude-opus-4-8**, five scoring repeats per pair and small anchor/partner queries (at most eight pairs). Extended thinking is disabled and only an ID-keyed numeric score JSON is requested. The initial batch is already frozen in `campaigns/hackathon2026/`; do not pay to regenerate it as part of normal setup.

What is cached in both stages

`hte/scoring.py` sends two stable text blocks with `cache_control={"type":"ephemeral","ttl":"1h"}`: (1) fixed chemistry, all individual ligand IDs/names/SMILES and available single-ligand SI yields, and (2) shared measured feedback. The short final block supplies one anchor and partner IDs. Cohorts/schema stay fixed across five repeats, while partner order is shuffled. The first actual scoring call warms each schema group; there is no extra paid warm-up. Shared feedback changes after experiments, so the second-round request has a new feedback prefix.

The live initial run reports **340 accepted requests, 311 cache-hit requests, 1,578,979 cache-read tokens and 147,721 cache-write tokens**. Accepted scoring cost is **$2.7599745** at the recorded rates. Prior rejected full-size output cost **$0.225235**, and earlier smoke tests **$0.4023195**: known combined estimate **$3.387529**. These are usage-based estimates, not an invoice. Saved counters and score audit.

Budget / recovery

The configured **$20 cap applies to each scoring campaign**, with `prior_campaign_cost_usd` carried into that campaign's ledger. It is not a global Anthropic account spending limit. Current mode is `incremental_cached`: before each request, reserve a conservative cold-cache amount, then replace that reservation with provider-reported usage only after a validated response. No SDK retries are enabled. Stop before the next reservation would exceed the cap. The theoretical all-miss initial bound was **$28.99**, so a complete $20 run depended on caching; actual accepted usage was $2.76. `strict_all_miss` mode instead rejects a campaign when its entire worst-case bound exceeds the cap.

`cost_preflight.json`, `budget_ledger.json`, per-request usage/raw responses and `cache_summary.json` remain with the run. A failed/uncertain request retains its reservation and attempt record; the pipeline will not silently repay it on restart. An incomplete score set cannot select a batch. No paid second-round calls have been made; real cost/latency may differ because experimental feedback enlarges the prompt. Initial scoring used two workers; feedback uses four. The five-minute second-round decision allowance is a target, not measured API throughput.

API key

Keep the raw key in `anthropic.key` **at the repository root**; copy the supplied `.example` file once and replace its placeholder. Restrict access with `chmod 600 anthropic.key`. Both initial scoring and feedback load it automatically. `ANTHROPIC_API_KEY` in the environment takes precedence. Keys are ignored by Git and excluded from the public handoff; provide HTE's key separately. Never add a key to robot Python or a configuration committed to GitHub.

Current official references: Anthropic caching, Opus 4.8. Recorded rates are $5/M input, $25/M output, 0.1× cache reads and 2× one-hour cache writes. They are configurable if pricing changes.

6. Software rehearsal evidence

Editable source: docs/rehearsal-results.md

Current machine-readable record: software_validation.json. Software checks validate implementation and simulation; no physical robot experiment, real calibration or 95 °C pair measurement has been performed.

Real initial Claude campaign

The real initial run is frozen in campaigns/hackathon2026. **435 pairs ×five repeats**, **340 accepted small anchor/partner requests**, all requested IDs validated, scores0-100, thinking disabled. Accepted usage estimate **$2.7599745**; 311 requests reported cache hits, 1,578,979 cache-read tokens. The 48-pair consensus covers **all30 active ligands**, with zero lower-quartile floor overrides. Exact accepted request, repeated scores, mean/SD ranking, source hashes and response IDs are saved. Initial scoring used two workers and took approximately9 min; this is not a measured feedback-loop latency.

Earlier full-size schema/array formats were rejected and never used for selection. Their costs and historical smoke tests are recorded in the audit; known combined estimated usage is$3.387529. No real second-round scoring has occurred.

Offline practice loop

`python -m hte.cli rehearse-loop --output output/practice` uses real frozen T5 vectors but mocks every Claude response and every chemistry yield. Current practice completed **650 mocked requests**, including340 initial requests and310 feedback requests for387 remaining pairs. Both stages contain **17 available individual SI yields**; each pair has five scores. All batches contain at most eight partners for one anchor.

The practice selects48 pairs covering30 ligands, accepts all48 synthetic LC results, then creates **22 untested pairs +L17 reference**. The complete second-round synthetic analysis also passes. It verifies no repeated pair, no SM redose and exactly one mesitylene dose after dry-down, ignores incomplete exports and repeated watcher events, and makes **zero paid API calls**. The generated practice next batch is not a scientific recommendation. No GoLLuM optimizer is invoked after feedback.

Acceptance scope

Tests cover stoichiometry/reserves/deck loads, supplied geometry and cap envelope, reaction-to-LC maps, acid/dilution arithmetic, ligand coverage, strict scoring, caching/budget stops and interrupted requests. Chromeleon batches with reset injection numbers require explicit maps and full coverage; partial/overlapping/duplicate exports are rejected. Calibration and quantitative QC are enforced.

Actual Opentrons8.8.2 simulations execute both rounds with custom labware, manual moves/cap pauses and tip replacement. Optional MOCCA and GoLLuM adapters are tested on synthetic data, without establishing performance for this chemistry.

Practical limits

Simulation does not establish physical fit, liquid temperature, extraction/IS recovery or LC separation. Formal separate homogeneity/solubility studies are not requested; pauses handle visible faults.

The 1.7-min full-cycle assumption and35-min handling overlap give717.7 min (752.7 min serial). No physical trial has demonstrated12 h. Record actual timings and remaining setup facts; public protocols retain physical-execution guards pending local release.